VEDANTA HAZRA

Mathematics and Computing (B.S. 4Y) — IIT Kharagpur

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2026	B.S. in Mathematics and Computing	IIT Kharagpur	8.48 / 10
2022	Class XII (ISC)	St. Xavier's Collegiate School	96.7%
2020	Class X (ICSE)	St. Xavier's Collegiate School	98.2%

PUBLICATIONS

Vedanta Hazra, Kaushal Kumar Bhagat. **Liquid-Gait: Bio-Adaptive AI for Mitigating Virtual Reality Cybersickness**

*Presented (Poster) at **IEEE VR 2026**, Daegu, South Korea | DOI: 10.1109/VRW70859.2026.00206*

- Designed a bio-adaptive controller using **Liquid Neural Networks (LNNs)** to modulate translation gain in real-time by fusing high-frequency ocular telemetry with irregular heart-rate variability, avoiding resampling artifacts associated with discrete AI approaches.
- Demonstrated a **46.8% reduction** in accumulated vestibular error and a **62.3% reduction** in saccadic jitter by actively regulating physiological arousal within a bounded limit cycle, suppressing sensory conflict compared with static mitigation baselines.

Vedanta Hazra, Kaushal Kumar Bhagat. **Semantic Dissonance Detection: A Contrastive Kinematic Transformer for Predictive Support**

*Accepted (Poster Presentation) at **IEEE ISMAR 2026**.*

- Developed a cross-modal safety framework that projects user intent (speech) and raw kinematic trajectories (OpenXR) into a shared **CLIP embedding space** to calculate real-time semantic cosine distance, reducing reliance on explicit object-level collision tracking.
- Achieved a **750ms predictive safety buffer** for identifying incongruent user actions prior to physical completion, enabling preemptive interventions that reduced completed motor errors by **95.3%** while maintaining a false-positive intervention rate of 1.2%.

Vedanta Hazra, Kaushal Kumar Bhagat, et al. **Gamified Virtual Tourism: Integrating Embodied AI and Spatial Play to Enhance Engagement**

*Accepted (Full Paper) at **IEEE ISEMV 2026**.*

- Engineered a gaze-grounded **Retrieval-Augmented Generation (RAG)** architecture and an asynchronous WebSocket streaming protocol to deploy a low-latency embodied conversational AI guide in a standalone VR environment, reducing perceived time-to-first-byte latency to 850ms.
- Designed a deterministic **Heuristic Dynamic Difficulty Adjustment (H-DDA)** engine for spatial physics puzzles, yielding a statistically significant improvement in objective knowledge recall (**79.4% vs. 54.2%**) and spatial presence without inducing cybersickness on mobile VR hardware.

RESEARCH & WORK EXPERIENCE

Member of Technical Staff — ORACLE — Bengaluru **July 2026 - Present**

- Engineering a fully **Redis-compliant server architecture** backed by an Oracle Database to bridge high-speed caching with enterprise relational storage.
- Developing system integrations for **True Cache** and **Data Guard**, alongside **SQL parsing optimizations**, to ensure high availability and minimal query latency.

Software Developer Intern (Research) — ORACLE — Bengaluru **May 2025 - July 2025**

- Constructed a real-time deadlock detection system using **Apache Kafka** and **Oracle PGX** to monitor complex database process-latch interactions.
- Deployed a scalable **Kafka Streams** pipeline and a multi-threaded processing solution that reduced graph analytics latency by **43%**.

Junior Researcher (Remote) — Deakin University, Australia — Supervisor: Dr. Gang Li **May 2024 - Sept 2024**

- Investigated privacy vulnerabilities in **Deep Reinforcement Learning (DRL)** algorithms, studying adversarial manipulation, model inversion, and membership inference attacks in multi-agent autonomous environments across diverse threat models and deployment settings.
- Developed adversarial attack methodologies to identify security gaps in autonomous agents, while implementing robust **adversarial training protocols** to improve DRL policy resilience against data poisoning during online deployment under realistic, continuously evolving operational conditions.

XR Developer — Evermorrow Labs — Pune **June 2024 - July 2024**

- Developed interactive **Mixed Reality (MR)** applications featuring precision gesture recognition and spatial audio integration.
- Optimized 3D assets and hardware interaction layers for standalone VR headsets to achieve sustained high frame rates and low-latency rendering.

KEY RESEARCH PROJECTS & ACHIEVEMENTS

EduScape XR (IEEE Metaverse Grand Challenge Winner) **Jan 2025 - Oct 2025** *First Prize Winner, IEEE Metaverse Grand Challenge 2025 — Presented at IEEE ISEMV (Hawaii)*

- Engineered a **modular VR framework using Unity and C#** that allows students to simulate STEM experiments by manipulating 3D objects and observing real-time physical or molecular interactions. Utilized Unity's physics engine, particle systems, and interaction toolkits to simulate accurate experiment behavior and outcomes
- Implemented intuitive **VR interactions and UI/UX** using hand tracking and spatial design principles for Oculus/Meta Quest devices. Created drag-and-drop interfaces for assembling circuits, mixing chemicals, and configuring experiment parameters, while optimizing performance for standalone VR headsets
- Designed an AI-driven pedagogical agent that analyzes user performance metrics to dynamically adapt simulation difficulty, improving learning outcomes.

Multi-Layer Spectral Ablation: Mapping and Mitigating Harmful Feature Restoration in Large Language Models **2026** *Unpublished Manuscript*

- Investigated how harmful representations can re-emerge across model layers after targeted feature ablation, identifying the layer-specific behavior and evolution of harmful feature restoration in large language models.
- Proposed a **multi-layer spectral intervention** that tracks changes in harmful feature subspaces and applies inference-time steering to improve the stability and effectiveness of model safety controls.

GameArt Nexus (AR Multiplayer Platform) **June 2024** *Finalist at JitHack Hackathon, Bangalore*

- Developed a multiplayer Augmented Reality battle environment using **Niantic Lightship SDK** and Unity, enabling shared AR experiences across multiple devices.
- Integrated blockchain protocols to facilitate secure, transparent asset transactions between graphic designers and gamers within the platform ecosystem.

BodySync Rush (Computer Vision Game) **Feb 2024**

- Developed BodySync Rush in **Unity 3D**, creating an immersive endless runner game where real-time body movements control the protagonist's actions. Implemented complex **game mechanics and physics** to enhance player engagement and interaction
- Utilized **OpenCV** and **MediaPipe** for low-latency pose estimation and skeletal tracking via standard webcams, eliminating the need for specialized hardware.

AWARDS AND SCHOLARSHIPS

- **1st Prize**, IEEE Metaverse Grand Challenge 2025 (Project: EduScape XR).
- **1st Prize**, XR Creator Hackathon, World Audio Visual and Entertainment Summit (WAVES) 2025.
- **Junior & Senior Scholar**, Jagadish Bose National Science Talent Search (JBNSTS) – Prestigious state-level science scholarship.
- **AIR 1210** (Top 0.7%), JEE Advanced 2022 among 172,000 candidates.
- **AIR 2909** (Top 0.2%), JEE Main 2022 among 1.2 million candidates.

LEADERSHIP

Research and Development Head — Computer Graphics Society, IIT Kharagpur **May 2024 - Present**

- Mentored student teams working on Computer Vision and Reinforcement Learning projects, guiding technical development and research direction.
- Led teams to victory in national VR Game Jams and the XR Creator Hackathon, coordinating development sprints and design strategy.